Discrete Mathematics (ST-1B)

1. Sets

A **set** is a collection of distinct objects.

Example:

A = {1,2,3,4}

Types of Sets

Type	Meaning
Empty Set	No elements
Finite Set	Limited elements
Infinite Set	Unlimited
Universal Set	All elements

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2. Set Operations

Union

$A \cup B$

All elements from both sets.

Example

A={1,2}
B={2,3}

$A \cup B = \{1,2,3\}$

Intersection

$A \cap B$

Common elements.

Example

$A \cap B = \{2\}$

Difference

$A - B$

Elements in A but not in B.

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3. Cartesian Product

$A \times B$

Example

$A=\{1,2\}$
$B=\{a,b\}$

$A \times B=\{(1,a),(1,b),(2,a),(2,b)\}$

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4. Relations

A relation is a **subset of Cartesian product**.

Example

$R=\{(1,2),(2,3)\}$

Types:

- Reflexive
- Symmetric
- Transitive